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Agenda

- **Brief about HTTP/2**
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- **What's gRPC ?**
- **gRPC Testing Tools**
- **gRPC Method Types**
- **gRPC Service Versioning**
- **Authentication and Authorization**
- **gRPC on .Net Supported Platforms**
- **gRPC-Web**
- **When to use gRPC ?**
- **gRPC JSON transcoding**

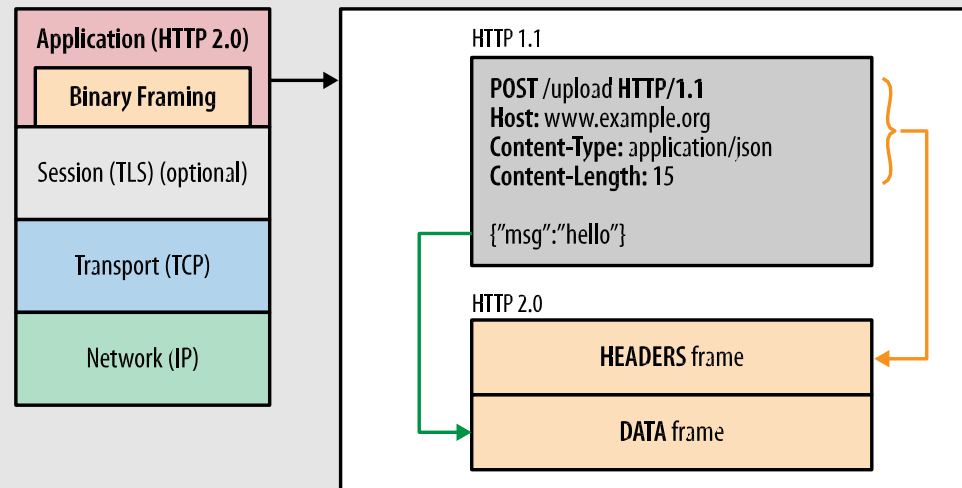


Brief about HTTP/2



- Experimental protocol called **SPDY** developed by **Google** & announced in 2009
- **SPDY** achieves improvement in page load to **55%** faster
- **HTTP/2** standard approved and published in 2015
- **HTTP/2**'s primary change focus on improving the performance:
- Semantics of **HTTP/1.x** are the same: methods, status code, URI and headers

- **Binary framing layer**

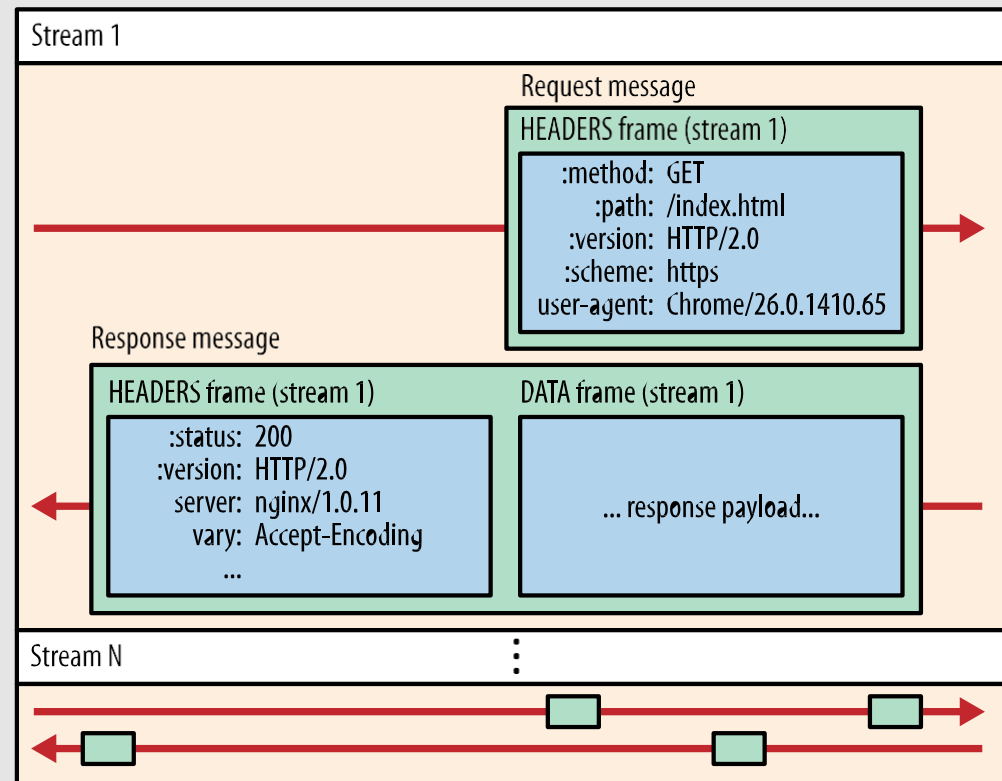


Brief about HTTP/2 (Cont.)



- Streams, messages and frames

Connection



Stream: A bidirectional flow of bytes within an established connection, which may carry one or more messages

Message: A complete sequence of frames that map to a logical request or response message

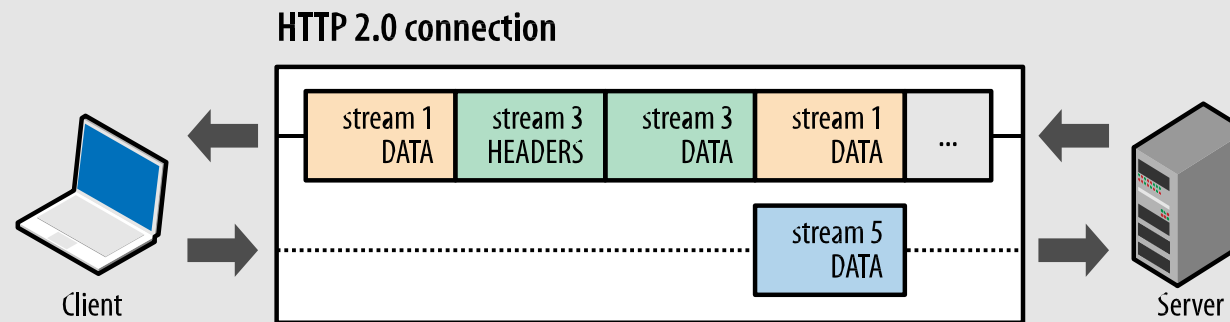
Frame: The smallest unit of communication, each containing a frame header identifies the stream to which the frame belongs.

All communication is performed over a **single TCP** connection that can carry **any number of bidirectional streams**.

Brief about HTTP/2 (Cont.)



- **Multiplexing**: multiple streams within same connection to enable parallel requests on single TCP connection
- **Server Push**: is the ability of the server to send multiple responses for a single client request



- **Header Compression** – HPACK Algorithm

gRPC Motivation

- General Purpose RPC infrastructure by **Google** called **Stubby**
- **Stubby** connect large number of microservices within **Google** DCs
- Not based on any standard

gRPC Principles

Free & Open

Layered

General Purpose &
Performant

Streaming

Payload Agnostic

Cancellation & Timeout

Blocking & Non-Blocking

Pluggable

Flow Control



What's gRPC ?



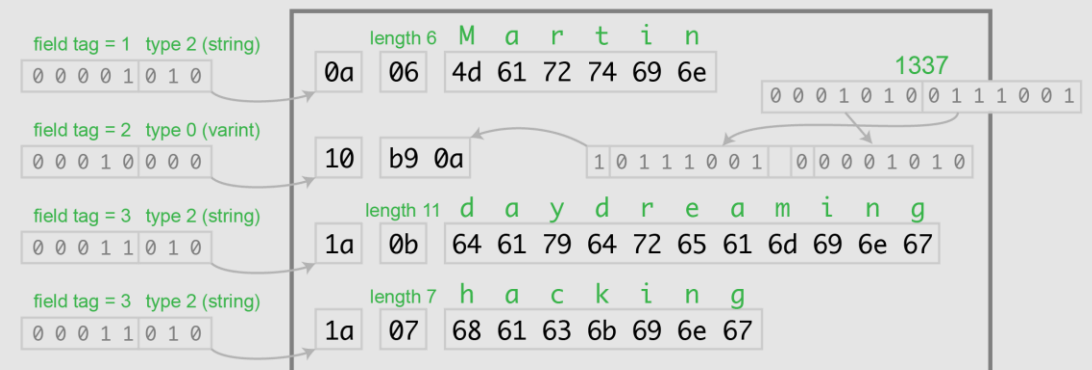
Microsoft | Docs:

gRPC is a language agnostic, high-performance Remote Procedure Call (RPC) framework.

Benefits

- High-performance, lightweight RPC framework
- By default, gRPC uses *Contract-first API* development using **Protocol Buffers** (aka **Protobuf**) as *Interface Definition Language*
- Reduced network usage with **Protobuf** binary serialization
- Supports client, server, and bi-directional streaming calls.

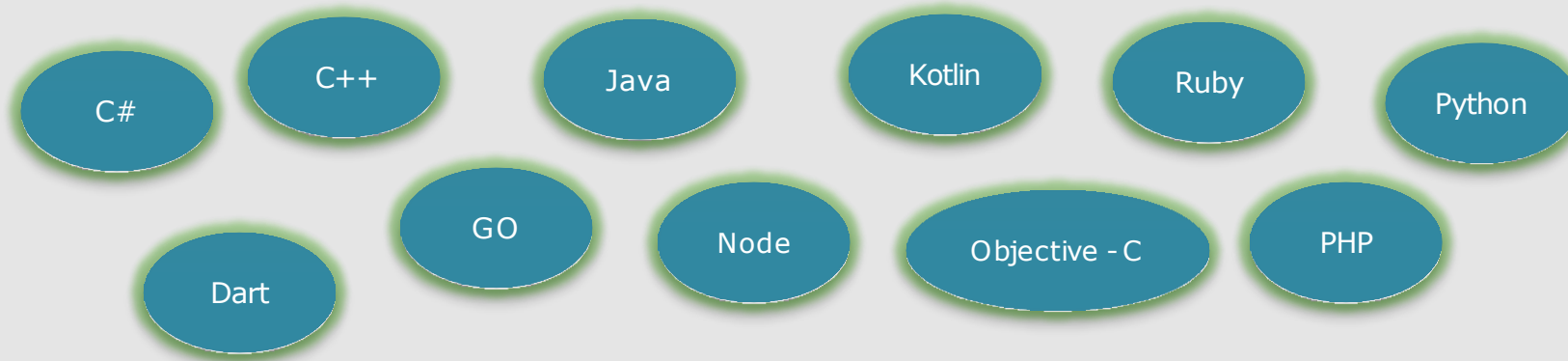
Protocol Buffers



total: 33 bytes

What's gRPC ? (Cont.)

- **gRPC** provides protocol buffer compiler plugins to generate strongly-typed servers and clients



Demo

ProtoBuf | My first Server | My first Client



gRPC Testing Tools

- **Postman:** under building blocks section
- **gRPCurl**
Command-line tool created by the gRPC community
- **gRPCui**
Interactive web UI for gRPC on top of gRPCurl
- **BloomRPC:** no support for gRPC reflection



gRPC Method Types

- **Unary**

```
rpc SayHello(HelloRequest) returns (HelloResponse);
```

- **Server Streaming**

```
rpc SayHello(HelloRequest) returns (stream HelloResponse);
```

- **Client Streaming**

```
rpc SayHello(stream HelloRequest) returns (HelloResponse);
```

- **Bi-directional streaming**

```
rpc SayHello(stream HelloRequest) returns (stream HelloResponse);
```



Demo

gRPCurl | gRPCui | Build Streams

